

# VAISHNAV AK

Data Scientist and AI Engineer

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## Professional Summary

Data Scientist and AI Engineer developing production LLM-powered agents, machine-learning forecasting, and cloud-native data pipelines for EV charging. Experienced in hybrid RAG, LLM tool calling, Python, SQL, and AWS across support automation, owner analytics, dynamic tariffs, customer retention, and operational monitoring. Earlier experience in AI consulting and biomedical engineering.

## Technical Skills

- **Programming & Databases:** Python, SQL, PostgreSQL, MongoDB, MySQL, Redis
- **Machine Learning:** Scikit-learn, XGBoost, Random Forest, Facebook Prophet, KMeans, TensorFlow, PyTorch, Keras, NLP, Computer Vision, Time-Series Forecasting
- **LLMs & AI Systems:** LLM Tool Calling, Hybrid RAG (BM25, ChromaDB, Cross-Encoder Reranking), Prompt Engineering, OpenAI API, AWS Bedrock (Claude), LangChain, LangGraph, CrewAI
- **Cloud & Data Engineering:** AWS (S3, EC2, Glue, Athena), PySpark, Parquet, ETL Pipelines, Flask, REST APIs
- **Analytics & Visualization:** Power BI, Pandas, Matplotlib, Seaborn, Statistical Modeling, Feature Engineering
- **Delivery & Reliability:** Docker, Git, Redis Caching, Regression Testing, Agile/Scrum, Technical Documentation

## Professional Experience

### Data Scientist and AI Engineer

Mar 2026 – Present

*chargeMOD*

*Thiruvananthapuram, Kerala*

Developed AI/ML and analytics solutions for EV infrastructure spanning **6,000+ chargers**, **250,000+ users**, and **3.5M+ transactions**.

### Customer Support AI Agent

- Reduced average support resolution time by **~50%** through a modular-monolith platform with three reasoning paths: direct log-based diagnosis, **BM25 + ChromaDB hybrid RAG** with cross-encoder reranking, and dynamic guided questioning.
- Implemented live API validation of charger IDs and wallet balances before LLM context injection for remote charger operations. Delivered fuzzy charger-name resolution and **SSE streaming** using OpenAI and AWS Bedrock (Claude), enabling new support staff to handle complex faults independently from day one.

### Dynamic Tariff & Demand Forecasting System

- Built a nightly two-level ML pricing pipeline with network-wide forecasting using **XGBoost**, **Random Forest**, and **Facebook Prophet**, automatic best-model selection, and per-charger pricing via an activity classifier paired with a log-space energy regressor.
- Drove **~10% revenue uplift** and peak-hour load suppression using **KMeans clustering of 36-feature charger profiles**, p95 outlier clipping, log1p transformation, and asymmetric sigmoid tariff adjustments within **-6% to +12%** limits.

### Client AI Agent – Natural Language Analytics

- Developed a conversational analytics agent serving **1,700+ daily active users** through **15–20 parameterised PostgreSQL query functions**, with owner ID injected from the session to enforce tenant-scoped access to energy, session, and fault data.
- Implemented Redis caching, nightly pre-aggregation, circuit breakers, per-owner rate limiting, and a deployment gate requiring at least **95% accuracy on prompt-regression tests**, reducing manual dashboard navigation.

Sales AI Agent

- Built a **CrewAI-based sequential workflow with four agents**—Site, Market, Financial, and Proposal Writer—to automate site feasibility, ROI calculations, and branded PDF proposals.
- Implemented strict Python tool boundaries for geolocation extraction, market metrics, and financial calculations.

Maintenance Tab – Uptime Monitoring

- Reduced operational downtime by ~20% through **AWS Glue/Python batch jobs** computing daily, gap-based uptime/downtime from MongoDB charger logs.
- Handled AC/DC schema differences, configurable gap thresholds, idempotent batch upserts, and area-wise drill-downs in the chargeMOD Console.

Customer Retention & Churn Reduction

- Built monthly DC-user churn detection using energy-drop thresholds and rolling-percentile lifecycle clustering across **five tiers**: Low, Medium, High, Very High, and Inactive.
- Delivered a proactive calling console, call-outcome logging, and **Power BI dashboards** tracking monthly active users, retention, and reactivation to support customer recovery.

Cloud ETL Pipeline & Power BI Dashboard

- Reduced cloud query costs by ~95% and ETL processing time by 30% using a nightly **AWS Glue + S3 Parquet + Athena** pipeline, transforming MongoDB transactions with PySpark and date partitioning.
- Implemented idempotent partition overwrites, a **48-hour lookback buffer**, and schema-anomaly quarantine; processed **1M+ transactions nightly** with daily Power BI refreshes.

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| <b>Biomedical Engineer Trainee</b>                                                                                                                                                                                                                                                         | Nov 2024 – May 2025      |
| <i>Aster Mother Hospital</i>                                                                                                                                                                                                                                                               | <i>Malappuram, India</i> |
| <ul style="list-style-type: none"><li>• Assisted with installation, calibration, and preventive/corrective maintenance of <b>100+ medical devices</b> in a NABH-accredited multi-specialty hospital; supported troubleshooting, safety testing, and hospital quality compliance.</li></ul> |                          |

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| <b>Data Science and AI Consultant Intern</b>                                                                                                                                                                                                                                                                                                                                      | Oct 2023 – Jun 2024     |
| <i>Rubixe</i>                                                                                                                                                                                                                                                                                                                                                                     | <i>Bangalore, India</i> |
| <ul style="list-style-type: none"><li>• Developed AI proof-of-concept projects and machine-learning models for retail and healthcare use cases using <b>Python, Pandas, and Scikit-learn</b>.</li><li>• Performed data preprocessing and feature engineering to improve model performance and training efficiency, translating ML algorithms into deployable solutions.</li></ul> |                         |

Education

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|----------------------------------------------------------|---------------------|
| <b>B.Tech in Applied Electronics and Instrumentation</b> | Jul 2019 – Jun 2023 |
| Government Engineering College Kozhikode   CGPA: 6.88    |                     |
| <b>Higher Secondary Education – Bio-Maths</b>            | Jul 2017 – Mar 2019 |
| G.H.S.S. Vazhakkad   91%                                 |                     |

Certifications & Courses

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|----------------------------------------------------------------------|---------------------|
| • <b>Certified Data Scientist</b> – IABAC; ID: IAB1120176000         | Jun 2025 – Jun 2028 |
| • <b>AI Expert</b> – IABAC; ID: IAB1120175905                        | May 2025            |
| • <b>Data Science Foundation</b> – IABAC; ID: IAB1120174667          | Oct 2024            |
| • <b>AI Expert</b> – DataMites, Professional Training                | Jul 2024 – Oct 2024 |
| • <b>Certified Data Scientist</b> – DataMites, Professional Training | Sep 2023 – Feb 2024 |
| • <b>Machine Learning</b> – Coursera; ID: WPDSNFXCRTWR               | Nov 2023            |
| • <b>MATLAB Workshop</b> – GEC Kozhikode & E-CELL IIT Bombay         | Jan 2021            |
| • <b>Python for Everybody</b> – Coursera; ID: M22GSCX3TWAU           | Nov 2020            |